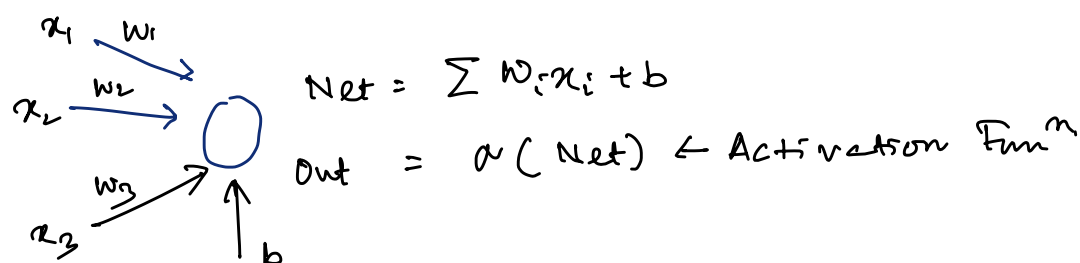
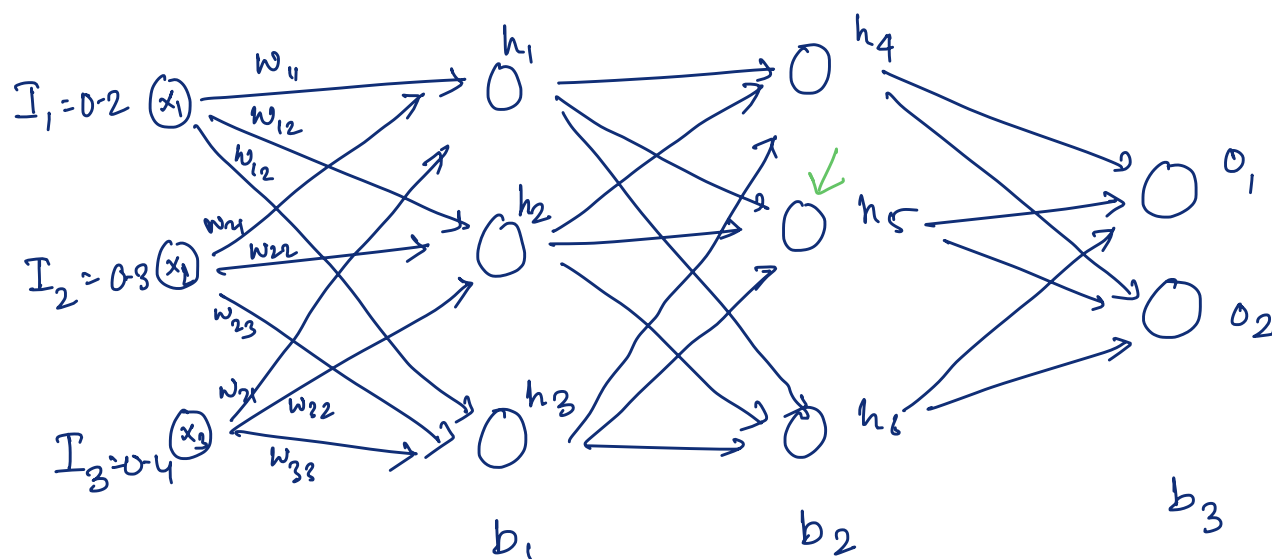


- history
- The Perceptron
- Multi-layer Perceptron
- Activation Functions

Back Propagation / Backprop.



Forward Pass $\rightarrow$ Error calculation $\rightarrow$ Backward Pass

Forward Pass

At h_1 ,

$$Net(h_1) = w_{11}x_1 + w_{21}x_2 + w_{31}x_3 + b_1 = 0.55$$

$$out(h_1) = \sigma(Net(h_1)) = \sigma(0.55) = 0.63$$

Same for h_2, h_3, h_4, h_5, h_6

Similarly

At o_1 :

$$Net(o_1) = out(h_4)w_{41} + out(h_5)w_{51} + out(h_6)w_{61} + b_2$$

$$out(o_1) = \sigma(Net(o_1))$$

Similarly for o_2

Error Calculation

loss:
Square loss

$$E_{total} = E(o_1) + E(o_2)$$

$$= \frac{1}{2} (actual_1 - out(o_1))^2 + \frac{1}{2} (actual_2 - out(o_2))^2$$

$$= 0.28$$

Backward Pass:

$$w_{41} \leftarrow w_{41} - \eta \Delta E_{total}$$

$$\leftarrow w_{41} - \eta \left(\frac{\partial E_{total}}{\partial w_{41}} \right)$$

$$w_{41} = 0.03$$

$$w_{41} = 0.03 + 0.001 \times 0.075$$

$$\frac{\partial E_{total}}{\partial w_{41}} = \frac{\partial E_{total}}{\partial out(o_1)} \cdot \frac{\partial out(o_1)}{\partial Net(o_1)} \cdot \frac{\partial Net(o_1)}{\partial w_{41}} = -0.075$$

$$\approx 0.0002$$

$$\frac{\partial E_{total}}{\partial out(o_1)} = \frac{\partial E(o_1)}{\partial out(o_1)} = \frac{\partial \left[\frac{1}{2} (actual_1 - out(o_1))^2 \right]}{\partial out(o_1)} = -(actual_1 - out(o_1)) = -0.01$$

$$\frac{\partial out(o_1)}{\partial Net(o_1)} = \frac{\partial (\sigma(Net(o_1)))}{\partial Net(o_1)} = \sigma(Net(o_1)) (1 - \sigma(Net(o_1)))$$

$$= out(o_1) (1 - out(o_1)) = 0.18$$

$$\frac{\partial Net(o_1)}{\partial w_{41}} = \frac{\partial (out(h_4) w_{41} + out(h_5) w_{51} + out(h_6) w_{61})}{\partial w_{41}} = out(h_4) = 0.69$$

Vanishing Gradient

$$w_{new} \leftarrow w_{old} - \eta \frac{\Delta E}{\Delta w}$$

- Skip Connection
- Gated Network
 - LSTM
 - GRU
- Attention - Transformer

Exploding Gradient

$$\left| \frac{dh}{dw} \right| < 1 \rightarrow \text{Vanishing Gradient}$$

$$\left| \frac{dh}{dw} \right| \geq 1 \rightarrow \text{Exploding Gradient}$$

Gradient Clipping

$t = \text{hyperparameter}$

$$\left| \frac{\partial h}{\partial w} \right| > t$$

$$\left| \frac{\partial h}{\partial w} \right| = t$$

$$g = \left| \frac{\partial h}{\partial w} \right| \leftarrow t \cdot \text{norm of } \left| \frac{\partial h}{\partial w} \right|$$

$$t = \frac{g}{\|g\|}$$

Weight Initialization

He Initialization / ReLU

Generate a random no.

→ betw n^{l-1}, n^l

np.random.rand
[n^{l-1}, n^l]

$$\times \sqrt{\frac{2}{n^{l-1}}}$$

n^l = # of neurons at l^{th} layer

n^{l-1} = " at $(l-1)^{th}$ layer

Xavier Initialization

tanh

np.random.rand [n^l, n^{l-1}]

$$① \sqrt{\frac{2}{n^{l-1} + n^l}}$$

$$② \sqrt{\frac{1}{n^{l-1}}}$$

Optimizers

Momentum

$$\Delta W_t = \alpha \Delta W_{t-1} + \eta \frac{\partial E}{\partial W_t}$$

$$W_{t+1} \leftarrow W_t + \eta \Delta W_t$$

$$\Delta W_t = (\beta) \Delta W_{t-1} - (1-\beta) \left(\frac{\partial E}{\partial W_t} \right)$$

→ momentum with moving Avg

RMS Prop

SGD fixed learning rate

→ Too large → the model oscillates or diverges

→ Too small → converges very slowly

scaling learning rate dynamically using moving average of past gradients

RMS Prop

$$MS(\Delta W_t) = \beta MS(\Delta W_{t-1}) + (1-\beta) \left(\frac{\partial E}{\partial W_t} \right)^2$$

$$W_{t+1} \leftarrow W_t - \eta \frac{\Delta W_t}{\sqrt{MS(\Delta W_t) + \epsilon}}$$

MS: Mean Square

ADAM (Adaptive Moment Estimator)

$$\Delta w_t = \beta_1 (\Delta w_{t-1}) + (1-\beta_1) \frac{\partial E}{\partial w_t}$$

$$\Rightarrow \Delta w_t = \frac{\Delta w_t}{1 - b_1^L} \quad b_1^L: \text{bias at 1st layer.} \quad \Rightarrow MS(\Delta w_t) = \frac{MS(\Delta w_t)}{(1 - b_2^L)}$$

$$w_{t+1} = w_t - \eta \frac{\Delta w_t}{\sqrt{MS(\Delta w_t) + \epsilon}}$$